

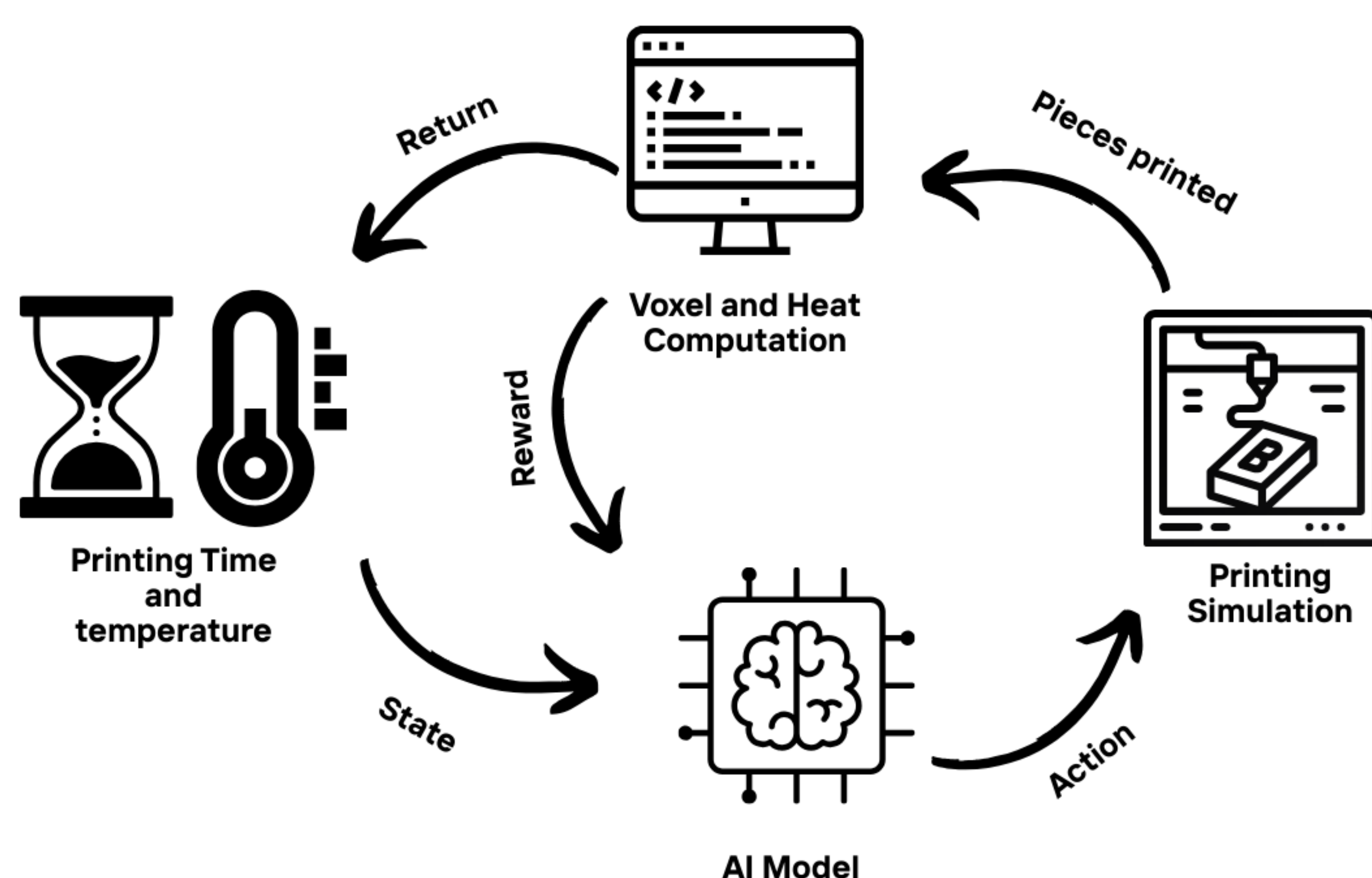
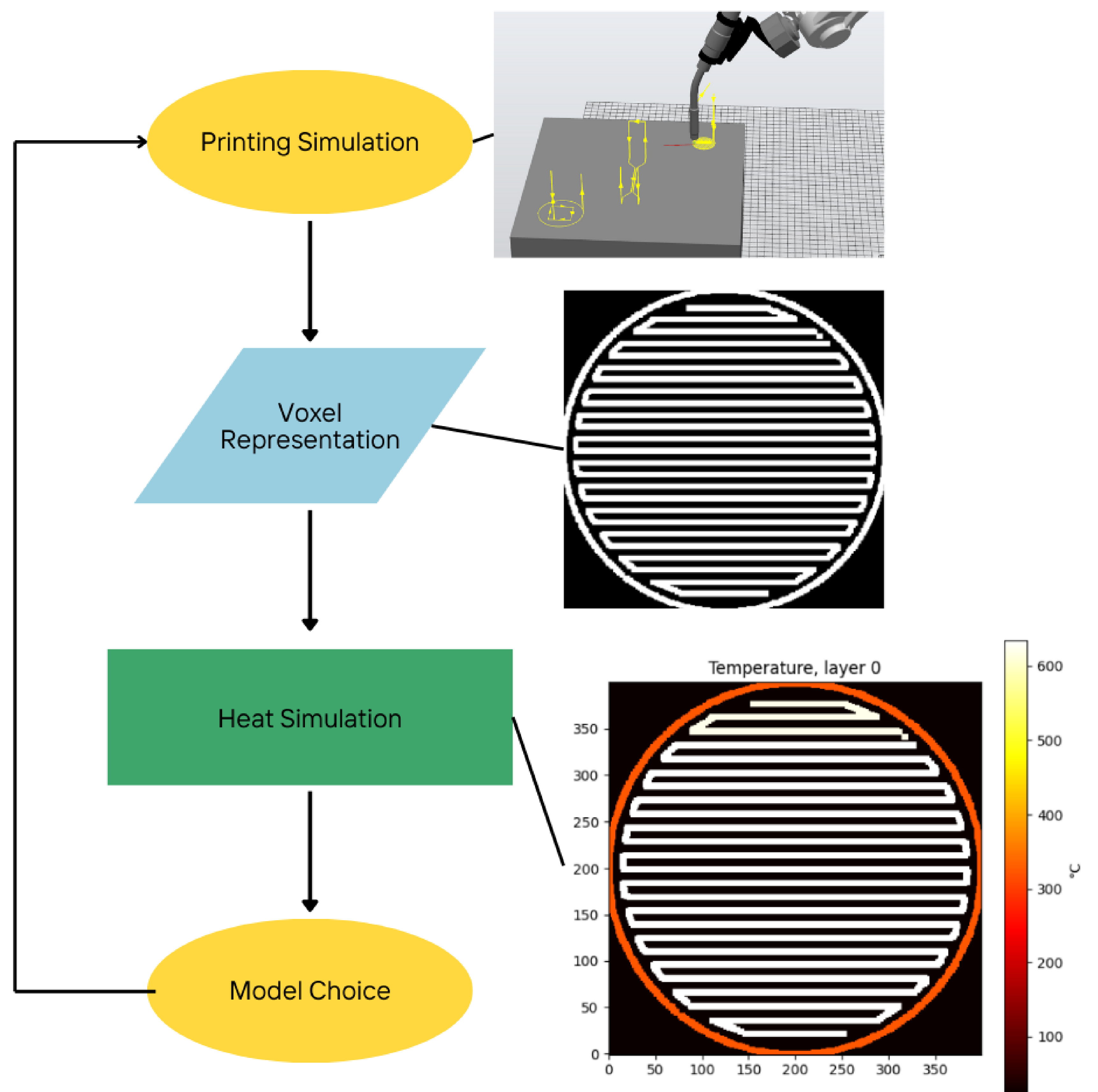
Improving Metal 3D Printing Efficiency with Heat Simulation and AI

Aims:

- Create a metal 3D printing simulation
- Create an accurate thermal simulation using voxel representation
- Implement AI decision making to optimize the printing sequence

Methodology:

- Robot Studio Simulation
- Voxel Based Geometry
- Heat Simulation
- Reinforcement learning implementation



Results:

- RL achieved up to 13% time saving compare to deterministic algorithm
- Successfully reduce waiting time in between pieces print
- Validate AI effectiveness across multiple scenarios

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